Statistical inference confidence intervals and hypotheses testing

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Outline

Definition

A statistical inference is a procedure that produces a probabilistic statement about some or all parts of a statistical model.

Some parts of a statistical model are not directly observable (e.g. parameters of probability distribution). Since the probability of any parameter estimate being completely correct is 0 (if the parametric space is uncountable), it is necessary to account for this via some probabilistic statements.

Confidence interval (CI)

Definition

Let $X_1, ..., X_n$ be IID random variables, that depend on a parameter θ and $g(\theta)$ be a real valued parametric function. Let $A \leq B$ be two statistics that have a property $P(A \leq g(\theta) \leq B) \geq (1 - \alpha)$. Then the random interval (A, B) is called $(1 - \alpha)$ confidence interval for $g(\theta)$.

Now if there is not enough observations of X to use any asymptotic properties, it is necessary to know the probability distribution of the estimator in question.
Example: Derive a non-asymptotic CI for μ from normal distribution with known variance (σ)

χ^2 distribution

Definition

Let $X_1, ...X_n$ be IID sample from standard normal distribution. Then $\sum_{i=1}^n X_i^2$ has the $\chi^2(n)$ distribution

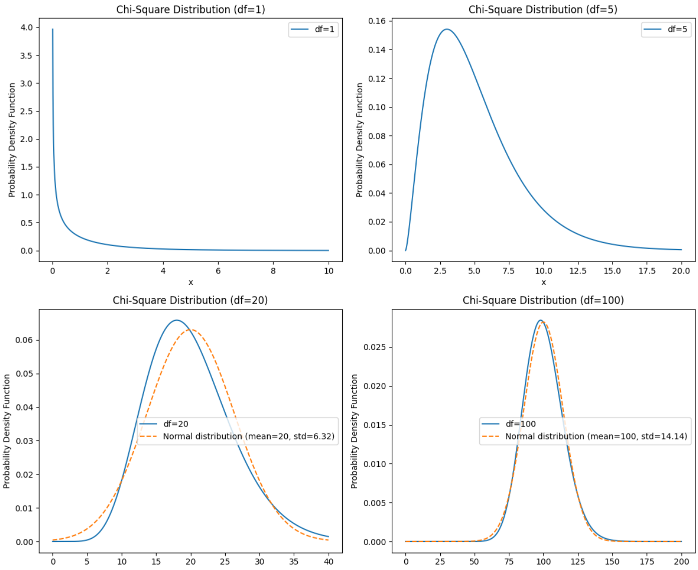
Theorem

If $X \sim \chi^2(n)$ then PDF of X is:

$$f(x) = \frac{1}{2^{n/2}\Gamma(n/2)} x^{n/2} e^{-x/2}; x > 0.$$

This corresponds to the Gamma distribution with parameters $\alpha = n/2$ and $\beta = 1/2$

Useful distributions



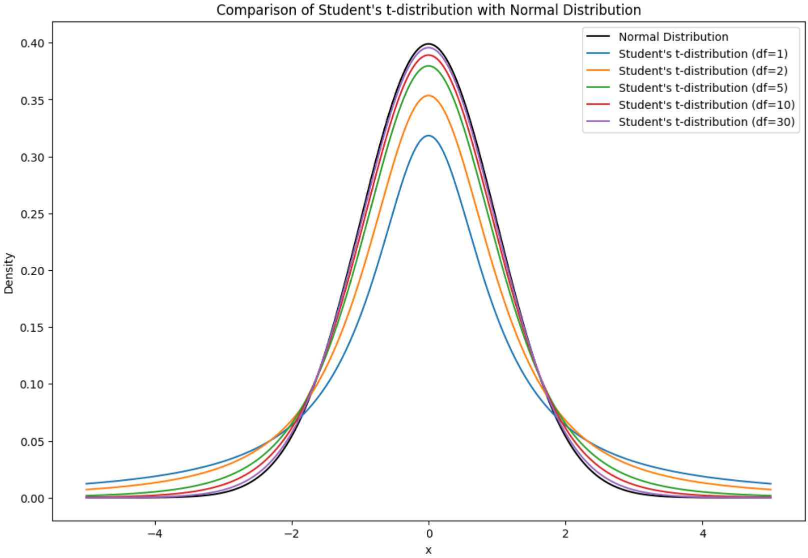
Student's t-distribution

Definition

Let Z be a random variable following standard normal distribution and Y be a random variable following $\chi^2(n)$ distribution. Let Z and Y be independent. Then

$$X = \frac{Z}{(Y/n)^{1/2}}$$

follows the t-distribution with n degrees of freedom.


Comparison of Student's t-distribution with Normal Distribution
Normal Distribution
Student's t-distribution (df=1)
Student's t-distribution (df=2)
Student's t-distribution (df=5)
Student's t-distribution (df=10)
Student's t-distribution (df=30)
Density
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−4
−2
0
2
4
x

CI's for parameters of normal distribution

Let $X_1, \dots X_n$ be IID normally distributed random variables, then CI's for σ^2 and μ are:

$$\sigma^2 \in \left(\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{\chi^2_{1-\alpha/2}}; \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{\chi^2_{\alpha/2}} \right)$$

$$\mu \in \left(\bar{X} - \frac{S(X)}{\sqrt{n}} t_{1-\alpha/2}; \bar{X} - \frac{S(X)}{\sqrt{n}} t_{1-\alpha/2} \right)$$

where $S(X) = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}$, $t_{1-\alpha/2}$ is $1 - \frac{\alpha}{2}$ quantile of t-distribution with $n-1$ degrees of freedom and $\chi^2_{1-\alpha/2}$, $\chi^2_{\alpha/2}$ are corresponding quantiles of χ^2 distribution with $n-1$ degrees of freedom.

Hypothesis testing

Hypothesis testing is a process used to make decisions about some statements regarding statistical model. e.g.:

- is the true value of a parameter θ of assumed probability distribution some known value θ_0
- do the observations follow assumed probability distribution
- are the values of a parameter θ of assumed probability distribution the same for 2 disjoint groups in the data

Generally speaking you want to decide if some statement regarding properties of a statistical model (that are not directly observable) is likely to be true.

Test for probability in a Binomial trials

Imagine you want to test a fairness of a classical die, mainly with regards to a probability of getting 6. However you can only observe results of dice throws. Find a way to decide whether a die is fair or not. Discuss possibilities for an errors in your conclusions.

Python illustration

Simple null hypothesis compounded alternative

Definition

If the hypothesis can be reduced to just single value (e.g. $H : \theta = \theta_0$) then we call the hypothesis simple, otherwise we call the hypothesis compounded.

Theorem

Under a simple hypothesis, the distribution of observations is fully specified. That means you can calculate the probability of observing any data, if the hypothesis is true.

How it works:

- Identify all necessary components of a statistical model.
- Decide on a sampling plan, and significance level α.
- "Translate" the problem into the null and alternative hypothesis.
- Observe the data.
- Use the data to compute a test statistic or p-value (use the probability distribution for the statistic, conditioned on H_0 being true)
- Considering resulting value, decide on rejecting or not-rejecting H_0

Definition

Let $X_1, ..., X_n$ be observed values from a distribution that depends on parameter θ, let $T = r(X_1, ..., X_n)$ be a statistic and let W be a subset of a real line. Suppose that the test procedure is in the form: "reject H_0 if $T \in R$" Then we call T test statistic (testové kritériun) and W the rejection region (kritický obor) of the test.

P-value

P-value is a probability of observing observed data, if the H_0 is true (and all assumptions of used statistical model hold). If the rejection region can be expressed from $T \geq c$ as an interval $\langle c; \infty)$, then the P-value is just a quantile function of the observed value of test statistic.

Errors in hypothesis testing

- Type I. error happens, if you reject a null hypothesis that is true.
  You set acceptable value for the probability of Type I. error (α) before the experiment, typically $\alpha = 5\%$.
- Type II. error happens, if you do not reject a null hypothesis that is not true.
  The probability of Type II. error (β) is a function of α, nuber of observations, selected test, variability etc.
- The Power of the test is the probability of rejecting a null hypothesis, if its false. $(1 - \beta)$

The t-test

Definition

Let $X_1, ..X_n$ be IID normally distributed random variables and it is required to test a hypothesis $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$. Then if H_0 is true, the following test statistic follows $t(n-1)$ probability distribution.

$$\frac{\bar{X} - \mu_0}{\sqrt{S^2(X)}}\sqrt{n}$$

.

To construct a rejection region it is necessary to get quantiles of t distribution. $W_\alpha = (-\infty; -t_{1-\alpha/2}) \cup (t_{1-\alpha/2}; \infty)$
Alternatively you can use a set complement to rejection region: $\bar{W}_\alpha = \langle -t_{1-\alpha/2}; t_{1-\alpha/2} \rangle$

One-sided alternatives

It can be useful to construct the alternative hypothesis in a way that it detects only that $\mu > \mu_0$ (or the other way around).
For the t-test, this can be handled just by adjusting rejection region.

If $H_0 : \mu = \mu_0$ with $H_1 : \mu > \mu_0$, then the rejection region will be $W_\alpha = (t_{1-\alpha}; \infty)$

If $H_0 : \mu = \mu_0$ with $H_1 : \mu < \mu_0$, then the rejection region will be $W_\alpha = (-\infty; -t_{1-\alpha})$